Total number of pages: 1 Roll no.:........................

Mid Semester Theory Examination,
Third Semester- B.Tech, September, 2025

Course code: COMTC13/CAMTC13/CMMTC13/CDMTC13
Course title: Probability and Stochastic Process

Time: 1 hour 30 minutes Maximum Marks. 25

Note: Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q. No.	Attempt all questions	Marks	CO
1a	A random variable X has the probability density function (pdf) $$f(x) = \begin{cases} \frac{1}{2}e^{-\lvert x\rvert}, & -\infty < x < \infty, \\ 0, & \text{otherwise.} \end{cases}$$ Find the mean and variance of the random variable X.	2.5	CO1
1b	The cumulative distribution function (CDF) of a continuous random variable X is defined by $F(X) = \begin{cases} 0, & x < 1; \\ A(x-1), & 1 \leq x \leq 4; \\ 1, & x > 4. \end{cases}$ Find (a) the value of A and (b) $\mathbb{P}(1 \leq X \leq 3)$.	2.5	CO1
2a	A random variable X has the following distribution: x: 0, 1, 2, 3, 4, 5, 6, 7 $P(x)$: 0, k, $2k$, $2k$, $3k$, k^2, $2k^2$, $7k^2+k$ (i) Find k. (ii) Evaluate $P(X < 6)$, $P(X \geq 6)$, and $P(0 < X < 5)$.	2.5	CO1
2b	The first four moments of a distribution about the value 5 are 2, 20, 40, 50. Find the kurtosis and the nature of the distribution curve.	2.5	CO1
3a	A probability curve $y = f(x)$ has a range from 0 to ∞. If $f(x) = e^{-x}$, find the mean and the third moment about the mean.	2.5	CO1
3b	A die is thrown 320 times and the following distribution of sixes is observed: No. of sixes: 0, 1, 2, 3, 4 Frequency: 220, 90, 10, 0, 0 Fit a binomial distribution and find the expected frequencies.	2.5	CO2
4a	Let X be a random variable following a Poisson distribution with mean 1. Show that $E[\lvert X-1\rvert] = \frac{2}{e}$.	2.5	CO2
4b	In the manufacture of car tyres, a particular production process is know to yield 10 tyres with defective walls in every batch of 100 tyres produced. From a production batch of 100 tyres, a sample of 4 is selected for testing to destruction. (a) Find the probability that the sample contains 1 defective tyre. (b) Find the expectation of the number of defectives in samples of size 4.	2.5	CO2
5a	Define probability density function(pdf) of Beta distribution of first kind, and find its mean and variance.	2.5	CO2
5b	If the random variable X is uniformly distributed over the interval $[-\sqrt{2}, \sqrt{2}]$, find the exact probabilities for $\mathbb{P}(\lvert X-\mu\rvert \geq \sqrt{\frac{3}{2}}\sigma)$. Further, use Chebyshev's inequality to find the lower bound of $\mathbb{P}(\lvert X-\mu\rvert < \sqrt{\frac{3}{2}}\sigma)$ and compare it with the exact probability.	2.5	CO2